Oxygen Delivery & Hypoxemia — Bedside Reference

Two levers: FiO_2 (most ward devices) and PEEP (HFNC small / CPAP / BiPAP / vent). Effective FiO_2 falls when inspiratory flow (30–120 L/min in distress) exceeds device flow. Adapted from TeachIM.org (Zhang, Oct 2021),

Device ladder (max effective FiO_2)

Device	Flow	Max FiO_2
Nasal cannula	2–6 L	~44% @6L (~3–4%/L)
Simple face mask	5–10 L	~60–70% (>6L washes CO_2)
Venturi	set	~60%, precise fixed FiO_2
Oxymizer	~12–15 L	>80% (reservoir)
Non-rebreather	12–15 L	~75–90% **RESCUE only**
HFNC	≤60 L	~100% + PEEP ~3–5 cm H_2O

Bigger 100%-O_2 reservoir = less entrained air: nasopharynx (~50 mL) < mask cup (100–250 mL) < NRB bag (600–1500 mL). HFNC floods the airway as its reservoir.

CPAP vs BiPAP

	CPAP	BiPAP
Pressure	Continuous (=PEEP)	IPAP + EPAP(=PEEP)
Affects	Oxygenation	Oxygenation + ventilation (ΔPAP→Vt)
Use	Hypoxemic	Hypoxemic + hypercapnic

Titrate: ↑FiO_2 or both pressures for O_2; widen ΔPAP (↑IPAP) to lower CO_2.

NIPPV indications & contraindications

Strong indications	Contraindications
Acute cardiogenic pulmonary edema (CPAP/BiPAP); COPD + hypercapnia (BiPAP); immunocompromised AHRF; post-op/post-extubation; palliative	AMS / can't protect airway; vomiting/copious secretions; facial trauma; recent esoph/tracheal surgery; hemodynamic instability

PPV hemodynamics: ↓RV preload, ↓LV preload, ↓LV afterload — unloads the heart in flash edema. **Not** for ARDS hypoxemia; weak in pneumonia unless concurrent COPD.

2022–25 framing

- **HFNC first-line** for non-cardiogenic/non-COPD AHRF (ESICM 2023, FLORALI NEJM 2015) — reduces intubation
- **ROX index** = (SpO_2/FiO_2)/RR; ≥4.88 reassuring, falling → escalate (Roca 2019)
- **Awake proning** adjunct in COVID AHRF — ↓ intubation (2023 guideline)
- CPAP ↓ intubation/death in COVID (RECOVERY-RS 2022); HFNC ≈ NIV overall

Targets & pearls

- SpO_2 goal 92–96% (88–92% if CO_2 retainer)
- NRB is rescue — de-escalate once stable; inflate the bag, need >10 L/min
- Venturi = best for a precise FiO_2 with variable breathing
- HFNC also washes dead space (↓CO_2 rebreathing) and ↓work of breathing